TOPIC 01 · THE FIRM

The Firm: Types and Objectives

What a business is, the circular flow of income, business functions and the management process, the entrepreneur, and the legal forms of business ownership with the agency problem.

Source: Universidad Carlos III de Madrid · Topic 1_The Firm_Types Objectives.pdf (Prof. Carmen Paz-Aparicio), Slides 1-29

OFFICIAL MATERIAL `Topic 1 The Firm.pdf` `W1 Elevator Pitch & Governance.pdf`

Central idea: a business is an economic activity that **transforms inputs into outputs** to satisfy people's needs. Firms and households exchange goods, services and factors of production in a **circular flow**; inside the firm, functions are organised and managed by an **entrepreneur** who holds the rights to the residual income; and the law offers several **forms of ownership** (sole proprietorship, partnership, corporation, cooperative, ...) whose differences in liability, taxation and permanence decide which one suits each business. In large corporations, owners and managers are different people, which creates the **agency problem**.

This note follows the official deck slide by slide. Every statement is taken from the slides; when the note adds an explanation of what a slide means, it only unpacks the slide's own words. Citations such as [Slide 5] open the PDF on that slide.

Contents of the topic [Slide 2]: (1) concept and nature of business; (2) business functional areas; (3) management process; (4) the role of engineers in business management; (5) the enterprise and the entrepreneur; (6) types of companies and legal forms.

§ 1 What a business is [Slide 3]

The slide answers the question "what is business?" with one definition:

DEFINITION OF BUSINESS [SLIDE 3]

Business is an **economic activity that transforms a set of inputs** (factors of production) **into outputs** (goods and services aimed to satisfy customers' needs and wants).

The two sides of that transformation are listed explicitly:

Inputs (factors of production)	Outputs
Land (natural resources)	Goods
Labour	Services
Capital	
Entrepreneurship	

Read the definition as a process: something comes in (the four factors), the business transforms it, and something more valuable goes out (goods or services that a customer wants). Entrepreneurship is counted as an input in its own right: someone has to bring the other three factors together (this idea returns in §7).

Example: the production process of bread [Slide 4]

The slide shows the **inputs** pictured inside an ellipse (a bag of sea salt, baking powder, wheat flour, sugar and a bottle of water), a large arrow with a **question mark**, and the **output**: loaves of bread. The question mark is the point of the exercise: the ingredients alone are not bread. Something must turn them into bread, and that "something" is the business activity of [Slide 3]: the land factor (the natural resources behind flour, salt and water), labour (the baker), capital (the oven and tools) and entrepreneurship (whoever organises it all) applied to the ingredients.

§ 2 The circular flow of income [Slides 5-11]

2.1 The model [Slide 5]

CIRCULAR FLOW OF INCOME [SLIDE 5]

The term **circular flow of income** refers to a simple economic model which describes the **reciprocal circulation of income between producers and consumers**. The continuous interaction of the markets of goods and services and of factors determines a **double flow, monetary and real**, through which the economy of a country decides **what to produce, how and to whom**.

The diagram of the slide has two agents, **households** and **firms**, joined by two markets:

- the **market of goods and services** (top loop): goods and services go from firms to households, and the households' **expenditures** go back to firms, where they are the firms' **revenues**;
- the **market of factors of production** (bottom loop): factors of production go from households to firms, and **wages, rents and dividends** go back to households.

What is left to the firm after paying for the factors is its **profit** (drawn as a separate output of the firm).

2.2 The two agents [Slides 6-8]

There are two main economic agents: **households** and **firms** [Slide 6].

Households play a double role [Slide 7]:

- on the one hand, they are the **owners of the production factors** (land, labour, capital, technology and "know-how");
- on the other hand, they are the **fundamental units of consumption**.

Firms use the production factors from households to produce the goods and services that households demand [Slide 8].

2.3 How they exchange [Slides 9-10]

The exchange between firms and households happens in markets:

- **Factor markets** [Slide 9]: households deliver inputs to firms and get incomes (wages, rents, dividends). This exchange takes place in the **markets of factors**, such as the **labour market** and the **stock-exchange market**.
- **Market of goods and services** [Slide 10]: firms produce goods and services that they exchange with households in this market, and firms get the money spent by households.

2.4 Real and financial flows [Slide 11]

In the economic system two types of flow can be distinguished:

- **Real flows**: goods, services and factors (what physically moves);
- **Financial flows**: their **monetary compensation** (the money that moves the other way).

Every real flow has a financial flow running in the opposite direction: goods go to households while expenditures go to firms; labour goes to firms while wages go to households.

§ 3 The nature of business: market and enterprise [Slide 12]

> ECONOMICS [SLIDE 12]
> Economics is the study of how people and a society choose to use **scarce resources** (factors of production) to satisfy people's **unlimited wants**.

The slide then asks how these needs can be satisfied, and answers with a chain:

1. Needs are satisfied through **specialization** = **division of labour** (each agent does what it does best).
2. Specialization requires **coordination** among the specialised agents.
3. Coordination can be carried out in two ways: by the **market** or by the **enterprise**.

The conclusion of the slide: **an enterprise comes up because there is a dissatisfied need**.

§ 4 Business functional areas [Slide 13]

The slide groups business functions into three blocks.

4.1 Real functions

The cycle drawn on the slide is **Inputs → Real activity → Outputs**. Its functions are:

- functions related to **supply** (acquisition of inputs);
- functions related to **production**;
- functions related to **marketing**;
- other **support** functions (Human Resources, Information Systems, ...).

4.2 Financial functions

Running parallel to the real cycle is the **financial activity**: the firm must **raise funds** (fundraising: search for financial resources), **pay for the acquisition of inputs**, collect the **revenues from sales**, and pay the **return of capital and interests** to those who provided the funds. Its functions are:

- functions related to **financing**;
- functions related to **investing**.

4.3 Management functions

- functions of **planning**;
- functions of **control**;
- functions of **organization**.

§ 5 The management process [Slide 14]

The management process consists of **planning, organizing, leading and controlling** (Henri Fayol, 1841-1925). The slide draws the four stages as a cycle around a common centre, **"achieving the organization's stated purpose"**, with one description per stage:

Stage	Description on the slide
Planning	Defining goals, establishing strategy, and developing subplans to coordinate activities.
Organizing	Determining what needs to be done, how it will be done, and who is to do it.
Leading	Directing and motivating all involved parties and resolving conflicts.
Controlling	Monitoring activities to ensure that they are accomplished as planned.

Because it is drawn as a cycle, controlling feeds back into planning: deviations found while monitoring lead to new goals and plans.

§ 6 The role of engineers in business management [Slide 15]

The slide lists what an engineer brings to business management:

- effective **project planning**, **time management** and **problem solving**;
- **attention to detail**;
- experience in **data processing**;
- **computer skills** and **technical knowledge**.

§ 7 The enterprise and the entrepreneur [Slides 16-18]

7.1 Definitions [Slide 16]

`ENTREPRENEUR AND ENTREPRENEURSHIP [SLIDE 16]`

An **entrepreneur** is a risk taker who starts and operates a business in hope of making a profit.
Entrepreneurship is the process of bringing land, labour and capital together and taking the risk involved in producing a good or service in the hope of making a profit.

Who is the entrepreneur? There is **no conclusive agreement** about who the entrepreneur is, nor about his/her role in the enterprise. The entrepreneur's duties are not clear, mainly in **big companies**, where there are owners, management directors, moneylenders, ... who seem to share some of the characteristics attributed to a single person: the entrepreneur [Slide 16].

7.2 Four ideas about who the entrepreneur is [Slide 17]

View	Author and year on the slide
Entrepreneur "Coordinator"	(no author given)
Entrepreneur "Risk Taker"	Knight, 1921
Entrepreneur "Innovator"	Schumpeter, 1950
Entrepreneur "On alert"	Kirzner, 1973

Therefore, the **entrepreneur's functions** are [Slide 17]:

- a person who **coordinates and controls** the productive process (coordinator);
- a person who **ensures the payment of the factors of production**;
- a person who **takes risky decisions** (risk taker);
- a person who **innovates** (innovator);
- a person who **detects new business opportunities** (on alert).

Each function matches one of the four views: the list is the union of them.

7.3 Internal organization of the enterprise [Slide 18]

The slide builds an argument in four steps.

Step 1. The internal organization of the enterprise is **teamwork**.

Step 2. Team production requires a CONTROLLER, someone who supervises the team.

Step 3. What is the **incentive** for performing this task? The controller gets four rights:

- to **receive the residual income or surplus** (what is left after paying all the other factors);
- to **control and reassign the other factors of production**;
- to be the **central party in all the contracts** signed with the inputs (raw materials, capital (K), human resources);
- to **sell the mentioned rights**.

Step 4. The entrepreneur is the person with all these rights.

So the entrepreneur is defined here not by a personality trait but by a bundle of rights: residual income, control, central contracting position and transferability of those rights.

§ 8 Types of enterprises: classification criteria [Slide 19]

Criterion	Classification
Economic: sector	Primary, secondary, tertiary
Economic: size	Big, medium and small
Economic: scope of the business	One market / multi-market; national / multinational
Legal: ownership of the factors of production	Private companies; state-owned companies
Legal: main forms of business ownership	Sole proprietorship; partnership; corporation

The slide highlights the last row (the **main forms of business ownership**): the rest of the topic develops each of them.

§ 9 Forms of business ownership [Slide 20]

The full map of forms of business ownership on the slide is:

- **Sole proprietorship**;
- **Partnership**: general partnership; limited liability partnership (LLP);
- **Corporation**;
- **Others**: cooperative associations; joint ventures; mutual companies, etc.

The following sections take them in the order of the deck.

§ 10 The sole proprietorship [Slide 21]

DEFINITION [SLIDE 21]

A **sole proprietorship** is a business owned by **one person**, who assumes all the risk, takes decisions and looks for business opportunities. Therefore, he/she **owns all the rights**.

Notice the link with §7.3: the sole proprietor holds every right of the entrepreneur at once.

Advantages	Disadvantages
1. Simple to start	1. Unlimited financial liability
2. Proprietor owns all profits	2. Hard to raise funds for expansion
3. Personal satisfaction (being your own boss)	3. Often has no one to share management responsibilities (one person's talents and skills)
4. Sole decision maker	4. Impermanence (it is not permanent)
5. No tax on business as distinct from owner	
6. Easy to dissolve	

Unlimited financial liability means that the owner answers for the debts of the business with his/her own wealth, because the business and the owner are not separate persons. "No tax on business as distinct from owner" means the same thing seen from the tax side: profits are taxed once, as the owner's income.

§ 11 The partnership [Slides 22-23]

DEFINITIONS [SLIDE 22]

A **partnership** is an association of **two or more persons** to carry on as **co-owners** of a business for profit. People form a partnership by entering into a **partnership agreement**.

A **partnership agreement** is a written contract between the owners of a partnership that identifies the business and states the partners' respective rights and duties (name, location, type of business, sharing of profits, dissolving the partnership, etc.).

Advantages [Slide 23]	Disadvantages [Slide 23]
1. Few restrictions on starting	1. Unlimited and joint financial liability
2. Pooling of funds, talents and borrowing power	2. Potential for personal disagreements
3. More chance to specialize than the sole proprietorship	3. Relative impermanence
4. No tax on business as distinct from owners	4. Frozen investment

Compared with the sole proprietorship, the partnership solves the "one person's talents" and "hard to raise funds" problems (several partners pool funds and skills), but liability becomes **unlimited and joint**: each partner can be made to answer for the debts of the whole business. A **frozen investment** means that a partner cannot easily take his/her money out of the business.

§ 12 The corporation [Slide 24]

DEFINITION [SLIDE 24]

A **corporation** is a legally chartered organization that is a **separate legal entity apart from its owners**. It has most of the rights that a person has.

Advantages	Disadvantages
1. Separate legal entity (shareholders are the owners)	1. Special and double taxation
2. Limited financial liability of owners	2. Complex and costly to form and dissolve
3. Permanence (it can exist forever)	3. Government regulation and public disclosure requirements
4. Easy transfer of ownership (shares)	
5. Greater financial capability	

Every advantage follows from being a **separate legal entity**: since the corporation itself owes its debts, the owners can lose only what they invested (limited liability); since it does not depend on any owner's life, it can last forever (permanence); since ownership is divided into shares, it can be transferred easily and many investors can contribute (greater financial capability). The price is **double taxation** (the corporation pays tax on its profit as a person distinct from its owners, and the owners are taxed again on what they receive), more cost and complexity, and regulation and disclosure.

Comparison of the three main forms (built from Slides 21-24)

Feature	Sole proprietorship	Partnership	Corporation
Owners	One person	Two or more co-owners	Shareholders
Liability	Unlimited	Unlimited and joint	Limited
Taxation	No tax on business as distinct from owner	No tax on business as distinct from owners	Special and double taxation
Permanence	Impermanent	Relatively impermanent	Permanent
Raising funds	Hard	Pooling of funds	Greater financial capability
Formation / dissolution	Simple / easy	Few restrictions	Complex and costly

§ 13 Separation of ownership and control: the agency problem [Slides 25-26]

13.1 The problem [Slide 25]

In **large corporations**, the **Board of Directors** guides the business. The slide describes the problem of separation of ownership and control:

- companies are managed by **managers**, who are **employees** and who take decisions and assume risks (they are professionals);
- managers just look for their **own benefit**, and the shareholder does not control their way of acting;
- managers may take decisions in their interest, **disregarding their effect on shareholders**;
- shareholders and managers have **different access to information** and **different objectives**;
- shareholders do not have enough information to assess the managers' performance, nor incentives to do it; they do not have incentives, either, to improve the management of the business.

The chain drawn on the slide is: **managers look for their own benefit → conflict of interest between ownership and control → agency problem**.

13.2 The solution [Slide 26]

Description of the solution: shareholders must be **able to control the managers' way of acting**, in order to avoid that they look for their own benefit. The **mechanisms of control** are:

1. **Selecting the Board of Directors**;

2. **Contracts signed with managers (goal linked)**: the manager's pay depends on reaching the goals that shareholders want;
3. **Requesting audits**;
4. **Additional mechanisms of control: indicators**.

The slide closes with a trend: **more and more, shareholders owning lots of shares are the ones taking decisions**.

§ 14 The cooperative [Slides 27-28]

DEFINITION [SLIDE 27]
A **cooperative association** is an incorporated organization whose **user-members (owners)** get back any revenue left after expenses are paid.

Principles [Slide 27]:

1. Membership is open.
2. Owned and democratically controlled by its members.
3. Economic benefits are distributed proportionally according to each member's level of participation in the cooperative.
4. Based on the values of self-help, self-responsibility, democracy and equality.
5. Cooperative members believe in the ethical values of honesty, openness, social responsibility and caring for others.

Examples [Slide 28]:

- **Employee credit unions**: they accept savings deposits from members who own shares in the co-op; members can borrow from the co-op; savers receive interest, and borrowers pay interest.
- **Agricultural co-ops**: they help member farmers market their products.
- **Buying co-ops**: they help members buy their products at lower prices.
- **Consumer co-ops**: they operate customer-owned retail facilities; consumers get together and form a buying pool to get quantity discounts and to replace the wholesalers and retailers.

The difference with the corporation is in principle 3: benefits are shared by **participation** (how much each member uses the cooperative), not by the capital each one contributed.

§ 15 Other forms of business ownership [Slide 29]

- A **limited partnership** is a business in which one or more, **but not all**, of the partners are liable for the firm's debts **only to the extent of their financial investment** in the firm.
- A **joint venture** is a special type of partnership set up by individuals or firms to **accomplish a specific task or project**. It is very important and usual in **international business**.

The limited partnership is a middle way between the partnership (all partners with unlimited liability) and the corporation (all owners with limited liability).

Study notes written from the official course material (Topic 1 The Firm.pdf; W1 Elevator Pitch & Governance.pdf). Online version: subjects/business-management/teoria/topic-1.html